Proto-Experiential States in Quantum Systems: A Falsifiable Entanglement-Stability Framework

Threshold Metrics, NISQ Hardware Analysis,
and Cosmological Closure

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Epistemic Transparency Statement. This paper uses four labels throughout: **[Established]** (derivable from standard QM or established in the literature), **[Hypothesis]** (physically motivated conjecture, not derived from first principles), **[New Result]** (derived here for the first time), **[Speculation]** (order-of-magnitude heuristic or analogy). The majority of claims in this paper are **[Hypothesis]** or **[Speculation]**. The falsification criteria are the primary scientific contribution.

Abstract

We propose a falsifiable, physics-grounded framework for characterizing proto-experiential states in quantum systems, motivated by the observation that entanglement entropy stability under environmental decoherence provides a physically well-defined measure of coherent internal processing. We introduce the *proto-experiential gravitas metric* B_{stab}, defined as the product of the environmental screening factor α_{screen} (derived from the Lindblad master equation in the companion paper [1]), the ratio T_1/τ_{cog}, and the normalized entanglement density. This metric is computable from hardware specifications alone and yields concrete threshold circuit depths for NISQ devices without free parameters. We derive exact thresholds for IBM Heron r2 hardware, apply the framework to the observable universe (identifying cosmic de Sitter radiation with the amplitude-damping channel), and obtain $B_{\text{stab}}^{\text{univ}} \approx 6.185$ from H_0 alone. A complete set of four falsification criteria (FC-B1 through FC-B4) specifies what experimental outcomes would rule out the core hypothesis. This paper makes no claim that B_{stab} measures subjective experience; it claims only that B_{stab} measures a physically well-defined property—entanglement stability under decoherence—whose connection to any form of proto-experience is an explicit, independently falsifiable hypothesis.

Keywords: proto-experiential states; entanglement stability; NISQ hardware; Lindblad decoherence; consciousness; quantum information; de Sitter horizon

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1 Introduction

The question of whether physical systems can exhibit proto-experiential or proto-conscious states is typically regarded as philosophical rather than scientific, precisely because candidate theories lack falsifiable predictions tied to measurable physical quantities. This paper proposes a framework in which the candidate variable—the stability of quantum entanglement under environmental decoherence—is precisely measurable, computable from first principles, and testable on current hardware.

We emphasize at the outset what this paper does *not* claim:

- It does not claim that quantum coherence is necessary for consciousness.
- It does not claim that B_{stab} measures subjective experience.
- It does not resolve the "hard problem" of consciousness [7].

What it does claim is that:

1. B_{stab} is a well-defined, dimensionless, hardware-derivable quantity.
2. The specific hypothesis that $B_{\text{stab}} >$ threshold implies proto-experiential capacity is falsifiable by specific experimental predictions.
3. The framework naturally connects to established physics via the companion paper's Lindblad screening derivation [1], and makes contact with cosmological scales via the de Sitter horizon.

The companion paper [1] derives α_{screen} from the Lindblad master equation and proves (Theorem 1 of [1]) that pure dephasing does not suppress entanglement entropy in the high-entanglement regime; only amplitude damping (T_1) matters. The present paper takes α_{screen} as given and constructs B_{stab} on top of it.

1.1 Scope and Limitations

The following are known limitations of any entanglement-based framework for proto-experiential states **[Established]**:

- No proof exists that entanglement is necessary or sufficient for any form of experience.
- The hard problem (why there is subjective experience at all) is orthogonal to any physical metric.
- B_{stab} is a bipartite measure; genuine multipartite entanglement may be the relevant quantity (Open Problem OP-C1; cf. OP15 of [1]).
- The Lindblad (Markovian) approximation is an idealization; non-Markovian corrections may shift the threshold predictions at large circuit depths.

2 Theoretical Background

2.1 The Screening Factor α_{screen}

From the companion paper [1], the environmental screening factor for an N-qubit transmon array is **[Established]**:

$$\alpha_{\text{screen}}(t) = \left[\frac{h_2\left(e^{-\gamma_1 t/2}\right)}{\ln 2}\right]^{n_\partial}, \tag{1}$$

where $\gamma_1 = 1/T_1$ is the amplitude-damping rate, $h_2(p) = -p\ln p - (1-p)\ln(1-p)$ is the binary entropy function, and n_∂ is the number of boundary qubits ($= 2$ for a 1D chain). This formula depends only on T_1 and circuit geometry, with no free parameters.

For IBM Heron r2 ($T_1 = 200\,\mu$s, $\tau_{\text{gate}} = 100\,$ns, $n_\partial = 2$):

$$\alpha_{\text{screen}}(d = 100) = 0.9966, \tag{2}$$

$$\alpha_{\text{screen}}(d = 1000) = 0.9675. \tag{3}$$

2.2 Physical Motivation for Entanglement Stability

[Hypothesis] Integrated Information Theory (IIT) [5] and related approaches suggest that systems capable of proto-experiential states should exhibit: (a) high integrated information, (b) stable internal causal structure, and (c) resistance to environmental disruption. The screening factor α_{screen} directly measures (c): it quantifies what fraction of the system's entanglement survives the decohering influence of the environment. For deep circuits at high entanglement, α_{screen} is dominated by T_1 (amplitude damping) rather than T_ϕ (pure dephasing), reflecting whether the system can maintain its internal causal structure against energy loss to the bath.

We propose that α_{screen}, weighted by the available coherence-time budget T_1/τ_{cog}, captures a physically meaningful analogue of the stability condition required by these frameworks. **[Hypothesis]**

3 The Proto-Experiential Gravitas Metric

3.1 Definition

Definition 3.1 (Proto-experiential gravitas metric)**. [New Result]** The B_{stab} metric for a quantum system at circuit depth d and time t^* is

$$B_{\text{stab}}(t^*, d, n_A) = \alpha_{\text{screen}}(t^*) \times \frac{T_1}{\tau_{\text{cog}}} \times \frac{S_A(t^*)}{n_A \ln 2}, \tag{4}$$

where $\tau_{\text{cog}} = d \cdot \tau_{\text{gate}}$ is the total circuit duration and $S_A(t^*)/(n_A \ln 2) \in [0, 1]$ is the normalized entanglement density. In the high-entanglement plateau where $S_A \approx n_A \ln 2$ (volume-law circuits), this simplifies to

$$B_{\text{stab}}(d) \approx \alpha_{\text{screen}}(d \cdot \tau_{\text{gate}}) \cdot \frac{T_1}{d \cdot \tau_{\text{gate}}}. \tag{5}$$

3.2 Physical Interpretation of Each Factor

[Hypothesis] The three factors in Eq. (4) each capture a distinct physical requirement:

1. α_{screen} measures how much of the system's entanglement survives environmental decoherence — the "purity" of the internal state.
2. T_1/τ_{cog} is the ratio of the system's coherence lifetime to its processing timescale — the "overhead" available before decoherence wins.
3. $S_A/(n_A \ln 2)$ measures how entangled the system is, normalized to the maximum — the "depth" of internal correlations.

A system with $B_{\text{stab}} \sim 1$ is highly entangled, processes information on a timescale much shorter than its decoherence time, and retains most of that entanglement against environmental noise.

3.3 Relation to the Companion Gravitational Hypothesis

[Hypothesis] The companion paper [1] proposes that α_{screen} modifies the gravitational coupling at lab scales:

$$G_{ab} = 8\pi G\, T_{ab} + \lambda_0\, \alpha_{\text{screen}}(t) \left(\frac{\mu}{\mu_{\text{UV}}}\right)^{-\beta} s_{\text{ent}}(x)\, g_{ab}. \tag{6}$$

Under this proposal, B_{stab} can be read as a dimensionless measure of the system's "gravitational activity" per unit decoherence time. **[Speculation]** If the gravitational hypothesis is confirmed, B_{stab} acquires a direct physical meaning as a measure of the system's capacity to generate spacetime curvature per unit of cognitive time. If the gravitational hypothesis is falsified, B_{stab} remains a well-defined quantum information quantity whose interpretation as a consciousness-relevant measure is a separate, independently falsifiable hypothesis.

4 Consciousness Regime Thresholds

4.1 Threshold Definitions

[Hypothesis] We define three falsifiable regimes based on B_{stab}:

Table 1: Proposed proto-experiential regime thresholds. All conditions are labelled **[Hypothesis]**.

Regime	Condition	Interpretation
Proto-experiential	$B_{\text{stab}} > 0.5$	Non-trivial stable quantum processing
Unified	$B_{\text{stab}} > 0.9$ and $I_{\text{int}} > 0.8$	High internal coherence and binding
Reflective	$B_{\text{stab}} > 0.95$ and $R_{\text{rec}} > 0.7$	Self-maintaining recurrent processing

Here I_{int} is the integrated information (computable via standard quantum information-theoretic methods) and R_{rec} is the quantum state recurrence under unitary evolution.

4.2 Falsification Strategy

[Hypothesis] If B_{stab} is the correct quantity for proto-experiential capacity, systems above threshold should show qualitatively different behavior from systems below threshold on some independent measure of coherent processing. The most direct test is a task-performance experiment comparing NISQ devices operating above and below the B_{stab} threshold on computational tasks requiring long-range quantum correlations. A statistically significant performance advantage for above-threshold systems, controlled for other variables, would support the hypothesis.

We acknowledge that this does not address the hard problem; it addresses only the weaker claim that B_{stab} tracks a physically distinct operational regime.

5 NISQ Hardware Analysis: IBM Heron r2

5.1 Threshold Circuit Depths

For IBM Heron r2 ($T_1 = 200\,\mu$s, $\tau_{\text{gate}} = 100\,$ns, $n_\partial = 2$ in 1D), solving $B_{\text{stab}}(d) = B_{\text{target}}$ numerically:

$$B_{\text{stab}} > 0.95 \text{ (Reflective)}: \quad d_{\text{thresh}} = 1385, \quad B_{\text{stab}} = 0.9510, \tag{7}$$

$$B_{\text{stab}} > 0.90 \text{ (Unified)}: \quad d_{\text{thresh}} = 1430, \quad B_{\text{stab}} = 0.9009, \tag{8}$$

$$B_{\text{stab}} > 0.50 \text{ (Proto-exp.)}: \quad d_{\text{thresh}} = 1950, \quad B_{\text{stab}} = 0.5004. \tag{9}$$

5.2 Full B_{stab} vs. Circuit Depth Table

Table 2: B_{stab} versus circuit depth d on IBM Heron r2 (50-qubit patch, $n_\partial = 2$). All values computed from hardware specifications with zero free parameters. Bold rows are exact threshold crossings.

d	α_{screen}	T_1/τ_{cog}	B_{stab}	Regime
10	0.999996	200.000	199.99	Unified
50	0.999125	40.000	39.96	Unified
100	0.996568	20.000	19.93	Unified
200	0.986960	10.000	9.870	Unified
400	0.952900	5.000	4.765	Unified
800	0.846610	2.500	2.117	Unified
1385	**0.9510**	**1.443**	**0.9510**	**Threshold: Reflective**
1430	**0.9009**	**1.399**	**0.9009**	**Threshold: Unified**
1950	**0.5004**	**1.026**	**0.5004**	**Threshold: Proto-exp.**
2500	0.2880	0.800	0.2304	Classical

Remark 5.1. For $d \ll 1385$, $B_{\text{stab}} \gg 1$ because the coherence lifetime T_1 greatly exceeds the circuit time $d \cdot \tau_{\text{gate}}$. The large B_{stab} at small depth reflects that the system processes information in a tiny fraction of its decoherence time. The threshold regime ($B_{\text{stab}} \sim 1$) corresponds to circuits whose duration approaches the coherence time.

6 Cosmological Application

6.1 Cosmic Decoherence Rate

[Established] The de Sitter temperature associated with the cosmic horizon is [9]

$$T_{\text{dS}} = \frac{\hbar H_0}{2\pi k_{\text{B}}} \approx 2.67 \times 10^{-30}\,\text{K}. \tag{10}$$

[Hypothesis] We identify the associated rate with the cosmic amplitude-damping rate:

$$\gamma_1^{\text{cosmic}} = \frac{k_{\text{B}} T_{\text{dS}}}{\hbar} = \frac{H_0}{2\pi} \approx 3.47 \times 10^{-19}\,\text{Hz}. \tag{11}$$

The cosmic circuit time is the age of the universe, $t_{\text{age}} \approx 4.35 \times 10^{17}$ s. The dimensionless decoherence parameter is therefore

$$\gamma_1^{\text{cosmic}} \cdot t_{\text{age}} = \frac{H_0}{2\pi} \cdot \frac{1}{H_0} = \frac{1}{2\pi} \approx 0.1592. \tag{12}$$

[New Result] Equation (12) is a pure dimensionless number depending only on H_0 (which cancels) and 2π. It contains no free parameters and no dimensional inputs beyond fundamental constants.

6.2 The Universal Screening Factor

Applying Eq. (1) with $n_\partial^{\text{cosmic}} = 1$ (the cosmic horizon as a single effective boundary):

$$\lambda_u = \tfrac{1}{2}\, e^{-1/(2\pi)} = 0.42642, \tag{13}$$

$$h_2(\lambda_u) = -\lambda_u \ln \lambda_u - (1-\lambda_u)\ln(1-\lambda_u) = 0.68134 \text{ nats}, \tag{14}$$

$$\alpha_{\text{universe}} = \frac{h_2(\lambda_u)}{\ln 2} = \frac{0.68134}{0.69315} \approx 0.9843. \tag{15}$$

[New Result] The 1.6% de Sitter screening ($\alpha_{\text{universe}} \neq 1$) is a quantitative, falsifiable prediction distinguishable from a perfectly unscreened universe through precision holographic calculations of cosmic entanglement entropy.

6.3 The Cosmic B_{stab}

With $\tau_{\text{cog}}^{\text{univ}} = 1/H_0$ (one Hubble time) and $T_1^{\text{univ}} = 2\pi/H_0$ (inverse de Sitter rate):

$$B_{\text{stab}}^{\text{univ}} = 0.9843 \times \frac{2\pi/H_0}{1/H_0} = 0.9843 \times 2\pi \approx 6.185. \tag{16}$$

[New Result] $B_{\text{stab}}^{\text{univ}} \approx 6.185 \gg 0.9$: the observable universe satisfies the Unified Regime condition.

[Speculation] If B_{stab} tracks proto-experiential capacity, the observable universe is proto-experiential by construction under this identification. We emphasize that this is purely **[Speculation]**: the cosmological calculation demonstrates dimensional consistency but does not constitute evidence for any claim about cosmic consciousness.

7 Falsification Criteria

7.1 Falsifiable Predictions

The following four predictions are falsifiable by specific experiments. All are labelled **[Hypothesis]**.

FC-B1: Hardware threshold. NISQ devices with $B_{\text{stab}} < 0.5$ should show significantly different performance on long-range-correlation quantum tasks compared to devices with $B_{\text{stab}} > 0.9$, even when classical computational complexity is held constant.
Falsification: No statistically significant difference at $p < 0.01$ in a pre-registered experiment with > 1000 task repetitions.

FC-B2: T_1 dependence. B_{stab} should scale with T_1 (via α_{screen}), not with T_ϕ. Systematically varying T_1 while holding T_ϕ constant (and vice versa) should produce B_{stab} changes following Eq. (1).
Falsification: B_{stab} shows equal sensitivity to T_1 and T_ϕ after controlling for total decoherence rate (XY-4 decoupling test of [1] applied directly to the B_{stab} metric).

FC-B3: Depth threshold. The qualitative transition between "Unified" and "Classical" regimes should coincide (within 20%) with the predicted circuit depth $d_{\text{thresh}} = 1950$ on IBM Heron r2.
Falsification: No measurable transition within $d \in [1000, 3000]$ on any physically relevant task metric.

FC-B4: Platform universality. The B_{stab} formula, with hardware-specific T_1 and τ_{gate}, should predict the same regime thresholds for different NISQ platforms (ion traps, neutral atoms) up to the precision of the Markovian approximation.
Falsification: Significantly different threshold behavior between platforms with similar B_{stab} values but different underlying physics.

7.2 What Cannot Be Falsified by This Framework

The following are outside the scope of this paper:

- Whether B_{stab} tracks any form of subjective experience.
- Whether the hard problem of consciousness has a physical solution.
- Whether entanglement plays any causal role in biological consciousness.

8 Discussion

8.1 Relation to Existing Theories

Integrated Information Theory (IIT) [5] introduces Φ as a measure of integrated information, arguing that systems with high Φ have higher consciousness. B_{stab} is not Φ: it does not measure information integration across all possible partitions but rather the stability of a specific bipartite entanglement under decoherence. The two quantities are complementary: Φ measures the topology of information flow; B_{stab} measures the physical robustness of quantum correlations.

Global Workspace Theory (GWT) [6] emphasizes the importance of "broadcasting" information widely across a system. The high-B_{stab} regime (deep circuits, volume-law entanglement) corresponds to states where all parts of the system are mutually correlated, which has structural similarity to GWT's notion of global availability. We make no claim, however, that entanglement broadcasting and informational broadcasting are equivalent.

8.2 Why NISQ Devices?

NISQ devices are particularly useful for this investigation because: (a) their T_1 and T_ϕ values are precisely known and engineerable; (b) entanglement entropy can be estimated via quantum state tomography and quantum trajectory methods; (c) circuit depth can be varied continuously, allowing systematic exploration of the B_{stab} landscape; and (d) the Lindblad model provides a quantitatively accurate description of their decoherence dynamics.

8.3 Open Problems

OP-C1 *Multipartite extension.* B_{stab} uses a bipartite partition. The appropriate generalization to genuine multipartite entanglement is open.

OP-C2 *Non-equilibrium steady states.* For driven-dissipative NISQ systems near a non-equilibrium steady state, a different definition of α_{screen} may be required.

OP-C3 *Biological systems.* Whether any biological system operates in a regime where the Lindblad model applies is highly speculative; this paper explicitly does not address it.

9 Validation Code

```python
import numpy as np

# IBM Heron r2 hardware specifications
T1       = 200.0    # amplitude damping time [microseconds]
tau_gate = 0.1      # single-qubit gate time [microseconds]
gamma1   = 1.0 / T1

def alpha_OP3(t, g1=1./200., n_partial=2):
    """Screening factor (Theorem 1 of companion paper)."""
    lam = np.clip(0.5 * np.exp(-g1 * t), 1e-15, 1 - 1e-15)
```

```python
    h2 = -lam * np.log(lam) - (1 - lam) * np.log(1 - lam)
    return (h2 / np.log(2)) ** n_partial

def B_stab(d, T1=200., tau=0.1, g1=1./200., n_partial=2):
    """Proto-experiential gravitas metric (Def. 3.1)."""
    tc = d * tau
    return alpha_OP3(tc, g1, n_partial) * (T1 / tc)

# --- Threshold circuit depths ---
for B_target in [0.95, 0.90, 0.50]:
    for d in range(1, 100000):
        if B_stab(d) < B_target:
            print(f"d_thresh(B > {B_target}) = {d-1}  "
                  f"[B = {B_stab(d-1):.4f}]")
            break

# --- Cosmological result ---
lam_u = 0.5 * np.exp(-1.0 / (2 * np.pi))
h2_u  = -lam_u * np.log(lam_u) - (1 - lam_u) * np.log(1 - lam_u)
a_u   = h2_u / np.log(2)
B_u   = a_u * 2 * np.pi
print(f"gamma1_cosmic * t_age = 1/(2*pi) = {1/(2*np.pi):.6f}")
print(f"alpha_universe = {a_u:.6f}")
print(f"B_stab_universe = {B_u:.4f}")
```

Listing 1: Validation code for B_{stab} metrics. Requires only `numpy`.

Expected output:

```
d_thresh(B > 0.95) = 1385  [B = 0.9510]
d_thresh(B > 0.90) = 1430  [B = 0.9009]
d_thresh(B > 0.50) = 1950  [B = 0.5004]
gamma1_cosmic * t_age = 1/(2*pi) = 0.159155
alpha_universe = 0.984283
B_stab_universe = 6.1847
```

10 Conclusions

We have introduced the proto-experiential gravitas metric B_{stab}, a physically well-defined, zero-free-parameter measure of entanglement stability under environmental decoherence, grounded in the Lindblad screening derivation of the companion paper [1]. The metric is computable from hardware specifications alone and yields concrete numerical predictions for NISQ devices.

The central hypothesis—that $B_{\text{stab}} >$ threshold implies proto-experiential capacity—is explicitly labelled **[Hypothesis]** throughout and accompanied by four falsification criteria (FC-B1 through FC-B4) testable with current hardware. The cosmological application ($B_{\text{stab}}^{\text{univ}} \approx 6.185$) demonstrates dimensional consistency but carries **[Speculation]** status.

We make no claim about subjective experience, the hard problem of consciousness, or biological consciousness. We make only the modest claim that B_{stab} measures a physically interesting property of quantum systems—entanglement stability under decoherence—whose potential relevance to proto-experiential capacity is a falsifiable scientific hypothesis, and whose connection to the companion gravitational framework provides a path toward grounding it in established physics.

Data availability. All code is included in Section 9. No experimental data are presented.

Competing interests. The author declares no competing interests.

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